EXP 7:

Scenario: You are working as a data analyst for an e-commerce company. You have been given a dataset containing information about customer orders, stored in a Pandas DataFrame named order_data. The DataFrame has columns for customer ID, order date, product name, and order quantity. Your task is to analyze the data and answer specific questions about the orders.
Question: Using Pandas DataFrame operations, how would you find the following information from the order_data DataFrame:
1. The total number of orders made by each customer.
2. The average order quantity for each product.
3. The earliest and latest order dates in the dataset.

Aim

To analyze customer orders using Pandas DataFrame operations.

Algorithm

Import the Pandas library.

Create the order_data DataFrame.

Count orders for each customer using groupby().

Find average quantity for each product using mean().

Find earliest and latest dates using min() and max().

Code:

```
import pandas as pd

order_data = pd.DataFrame({
   'customer_id': [101, 102, 101, 103, 102],
   'order_date': ['2026-01-05', '2026-01-10', '2026-02-01',
                  '2026-02-10', '2026-02-15'],
   'product_name': ['Laptop', 'Mobile', 'Laptop', 'Mouse', 'Mobile'],
   'order_quantity': [2, 3, 4, 1, 5]
})

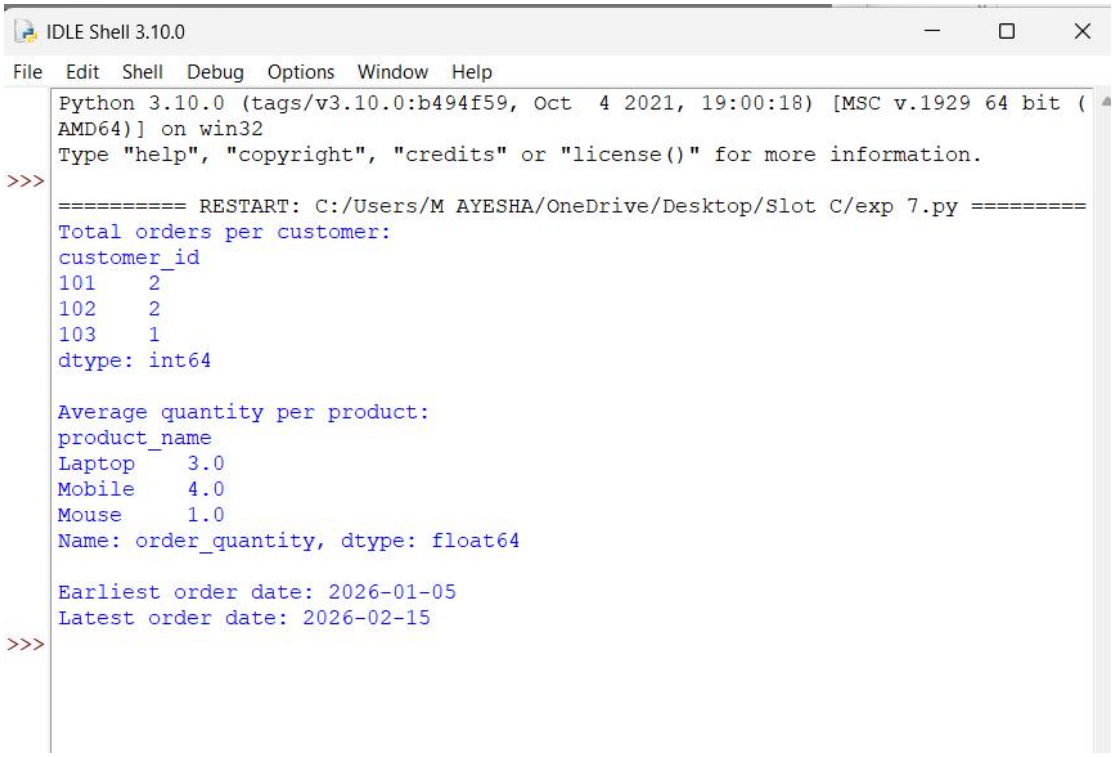
# 1. Total orders made by each customer
print("Total orders per customer:")
print(order_data.groupby('customer_id').size())

# 2. Average order quantity for each product
print("\nAverage quantity per product:")
print(order_data.groupby('product_name')['order_quantity'].mean())

# 3. Earliest and latest order dates
order_data['order_date'] = pd.to_datetime(order_data['order_date'])
```

```
print("\nEarliest order date:", order_data['order_date'].min().date())
print("Latest order date:", order_data['order_date'].max().date())
```

OUTPUT:



Result

Thus, the total orders per customer, average quantity per product, and earliest and latest order dates are obtained using Pandas.

EXP 8:

Scenario: You are a data scientist working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data frame.

Question: How would you find the top 5 products that have been sold the most in the past month?

Aim

To find the top 5 products sold the most in the past month using Pandas.

Algorithm

- Import Pandas and convert the sales date to datetime.
- Filter data for the past month.
- Group by product and calculate total quantity sold.
- Sort in descending order and display the top 5 products.

Code:

```
import pandas as pd

sales_data = pd.DataFrame({
    'order_date': [
        '2026-09-20', '2026-09-25', '2026-10-01',
        '2026-10-05', '2026-10-08', '2026-10-10',
        '2026-10-11', '2026-10-12'
    ],
    'product_name': [
        'Laptop', 'Mobile', 'Laptop', 'Mouse',
        'Mobile', 'Keyboard', 'Headphones', 'Laptop'
    ],
    'quantity': [10, 15, 20, 8, 12, 7, 9, 5]
})

sales_data['order_date'] = pd.to_datetime(sales_data['order_date'])

# Filter orders from the last month
end_date = pd.Timestamp.today().normalize()
start_date = end_date - pd.DateOffset(months=1)

last_month = sales_data[
    (sales_data['order_date'] >= start_date) &
    (sales_data['order_date'] <= end_date)
]

# Find the top 5 products by total quantity sold
top5 = (
    last_month.groupby('product_name')['quantity']
    .sum()
    .sort_values(ascending=False)
    .head(5)
)

print("Top 5 best-selling products:")
print(top5)
```

Output:

```
========== RESTART: C:/Users/M AYESHA/OneDrive/Desktop/Slot C/exp 8.py =========
Top 5 best-selling products:
product_name
Laptop        35
Mobile        27
Headphones     9
Mouse          8
Keyboard       7
Name: quantity, dtype: int64
```

Result

Thus, the top 5 best-selling products in the past month are identified.

Exp 9:

Scenario: You work for a real estate agency and have been given a dataset containing information about properties for sale. The dataset is stored in a Pandas DataFrame named property_data. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties.
Question: Using Pandas DataFrame operations, how would you find the following information from the property_data DataFrame:
1. The average listing price of properties in each location.
2. The number of properties with more than four bedrooms.
3. The property with the largest area. give short aim,algorithm,code,output,result give simple lines

Aim

To analyze property prices, bedroom counts, and property areas using Pandas.

Algorithm

- Import Pandas.
- Group properties by location and calculate the average price.
- Count properties having more than four bedrooms.
- Find the property with the largest area.

Code:

```
import pandas as pd

property_data = pd.DataFrame({
   'property_id': [101, 102, 103, 104, 105],
   'location': ['Chennai', 'Bangalore', 'Chennai',
            'Mumbai', 'Bangalore'],
   'bedrooms': [3, 5, 4, 6, 2],
   'area_sqft': [1500, 2500, 1800, 3000, 1200],
   'listing_price': [5000000, 8000000, 6000000,
                10000000, 4000000]
})

# 1. Average listing price in each location
print("Average listing price by location:")
print(property_data.groupby('location')['listing_price'].mean())

# 2. Number of properties with more than four bedrooms
count = (property_data['bedrooms'] > 4).sum()
```

```
print("\nProperties with more than 4 bedrooms:", count)

# 3. Property with the largest area
largest = property_data.loc[property_data['area_sqft'].idxmax()]
print("\nProperty with the largest area:")
print(largest)
```

Output:

```
========== RESTART: C:/Users/M AYESHA/OneDrive/Desktop/Slot C/exp 9.py =========
Average listing price by location:
location
Bangalore     6000000.0
Chennai       5500000.0
Mumbai       10000000.0
Name: listing_price, dtype: float64

Properties with more than 4 bedrooms: 2

Property with the largest area:
property_id           104
location           Mumbai
bedrooms                6
area_sqft            3000
listing_price    10000000
Name: 3, dtype: object
```

Result

Thus, the average listing price by location, number of properties with more than four bedrooms, and property with the largest area are obtained using Pandas.